

UNIVERSITÀ DEGLI STUDI DI TORINO

Corso di Laurea Magistrale in Matematica



UNIVERSITÀ
DI TORINO

Tesi di Laurea Magistrale

Teoria delle categorie

Relatore

professor Alan Cigoli

Candidato

Christian D'Ascoli

Anno Accademico 2024-2025

Indice

1	Teoria	3
1.1	The pullback functor	3
	Bibliografia	9

Introduzione

Testo

Capitolo 1

Teoria

1.1 The pullback functor

In this section we will define the pullback functor and its left adjoint, the shrink functor.

Given a category $\mathcal{A}$ with pullbacks and a morphism $p : E \rightarrow A$ in $\mathcal{A}$, one can consider adjunction

$$(\mathcal{A} \downarrow B) \quad \begin{array}{c} \xleftarrow{p^*} \\ \perp \\ \xrightarrow{p!} \end{array} \quad (\mathcal{A} \downarrow E)$$

defined as follows:

- $(\mathcal{A} \downarrow B)$ is the category of objects over B in $\mathcal{A}$, we denote an object by $(A, \alpha : A \rightarrow B)$;
- $(\mathcal{A} \downarrow E)$ is the category of objects over E in $\mathcal{A}$, we denote an object by $(D, \delta : D \rightarrow E)$;
- $p^* : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$ is the pullback functor

$$\begin{array}{ccc} \begin{array}{c} A \\ \downarrow \alpha \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_2} & A \\ \downarrow \text{proj}_1 & & \downarrow \alpha \\ E & \xrightarrow{p} & B \end{array} \end{array}$$

$$\begin{array}{ccc}
 \begin{array}{ccc} A & \xrightarrow{f} & A' \\ \downarrow \alpha & \searrow \alpha' & \\ B & & \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \swarrow & \text{dashed} & \searrow & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}'_2} & E \times_B A' \\ & \searrow \alpha & & \swarrow \alpha' & & \\ & & B & & & \\ \text{proj}_1 \swarrow & & \uparrow p & & \searrow \text{proj}'_1 & \\ & & E & & \end{array} \end{array}
 \end{array}$$

- $p_! : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is the shrink functor

$$\begin{array}{ccc}
 \begin{array}{ccc} D & & \\ \downarrow \delta & & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & & \\ \delta \downarrow & \searrow p\delta & \\ E & \xrightarrow{p} & B \end{array} \\
 \\
 \begin{array}{ccc} D & \xrightarrow{d} & D' \\ \delta \downarrow & \searrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{d} & D' \\ \delta \downarrow & \searrow p\delta & \downarrow p\delta' \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$

- the unit η^p

$$\begin{array}{l}
 \eta : \text{Id}_{(\mathcal{A} \downarrow E)} \rightarrow p^* p_! \\
 \eta_{(D, \delta)} = \langle \delta, 1_D \rangle : (D, \delta) \rightarrow (E \times_B D, \text{proj}_1)
 \end{array}$$

$$\begin{array}{ccccc}
 D & & \xrightarrow{1_D} & & D \\
 \delta \downarrow & \searrow \exists! \langle \delta, 1_D \rangle & & \searrow & \downarrow p\delta \\
 E \times_B D & \xrightarrow{\quad} & D & & \\
 \downarrow \text{proj}_1 & & \downarrow p & & \\
 E & \xrightarrow{p} & B & &
 \end{array}$$

- the counit ϵ^p

$$\begin{array}{l}
 \epsilon : p_! p^* \rightarrow \text{Id}_{(\mathcal{A} \downarrow B)} \\
 \epsilon_{(A, \alpha)} = \text{proj}_2 : (E \times_B A, p(\text{proj}_1)) \rightarrow (A, \alpha)
 \end{array}$$

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\text{proj}_2} & A \\
 \downarrow \text{proj}_1 & & \downarrow \alpha \\
 E & \xrightarrow{p} & B
 \end{array}$$

Let us show that this is actually an adjunction. First we prove that the unit is a natural transformation. Let $g : (D, \delta) \rightarrow ((D', \delta'))$ be a morphism in $(\mathcal{A} \downarrow E)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D \\
 \downarrow g & & \downarrow p^*p_!(g)=\tilde{g} \\
 D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D'
 \end{array}$$

In order to do that we need to expand the action of the functor $p^*p_!$ on the morphism g . We have:

$$\begin{array}{ccc}
 \begin{array}{ccc}
 D & \xrightarrow{g} & D' \\
 \downarrow \delta & \searrow \delta' & \\
 E & &
 \end{array} & \xrightarrow{p_!} & \begin{array}{ccc}
 D & \xrightarrow{g} & D' \\
 \searrow p\delta & & \downarrow p\delta' \\
 & & B
 \end{array} \\
 \\
 \xrightarrow{p^*} & & \begin{array}{c}
 \exists! \tilde{g} \\
 \curvearrowright \\
 \begin{array}{ccccc}
 E \times_B D & \xrightarrow{\text{proj}_2} & D & \xrightarrow{g} & D' & \xleftarrow{\text{proj}'_2} & E \times_B D' \\
 & \searrow \eta_{(D,\delta)} & \downarrow p\delta & \downarrow p\delta' & \searrow \eta_{(D',\delta')} & & \\
 & & B & & & & \\
 \text{proj}_1 \swarrow & & \uparrow p & & \swarrow \text{proj}'_1 & & \\
 & E & & & & &
 \end{array}
 \end{array}
 \end{array}$$

Considering the diagram

$$\begin{array}{ccccc}
 D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & & \\
 \downarrow g & \searrow \tilde{g} & \downarrow p^*p_!(g)=\tilde{g} & & \\
 D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\
 & & \downarrow \text{proj}'_1 & & \downarrow p\delta' \\
 & & E & \xrightarrow{p} & B
 \end{array}$$

we can prove the naturality in the left diagram by the universal property of the pullback in the right. Since $(\text{proj}_1, \text{proj}_2)$ are jointly monic, by the following compositions

- $(\text{proj}'_2)(\tilde{g})(\eta_{D,\delta}) = (g)(\text{proj}_2)(\eta_{D,\delta}) = (g)(1_D) = g$
- $(\text{proj}'_2)(\eta_{D',\delta'})(g) = (1_{D'})(g) = g$
- $(\text{proj}'_1)(\tilde{g})(\eta_{D,\delta}) = (\text{proj}_1)(\eta_{D,\delta}) = \delta$
- $(\text{proj}'_1)(\eta_{D',\delta'})(g) = (\delta')(f) = \delta$

we conclude that $(\tilde{g})(\eta_{(D,\delta)}) = (\eta_{(D',\delta')})(g)$. So the naturality condition of η is satisfied.

Now we want to prove the counit is a natural trasfromation. Consider $f : A \rightarrow A'$ a morphism in $(\mathcal{A} \downarrow B)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)}} & a \\
 p!p^*(f)=\tilde{f} \downarrow & & \downarrow f \\
 E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')}} & A'
 \end{array}$$

Considering the of the functor $p!p^*$ on the morphism f , we have:

$$\begin{array}{ccc}
 \begin{array}{ccc}
 A & \xrightarrow{f} & A' \\
 \downarrow \alpha & \nearrow \alpha' & \\
 B & &
 \end{array} & \xrightarrow{p^*} & \begin{array}{ccccc}
 & & \exists! \tilde{f} & & \\
 & \text{---} \text{dashed} \text{---} & & \text{---} \text{dashed} \text{---} & \\
 E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}'_2} & E \times_B A' \\
 & \searrow \alpha & \downarrow & \swarrow \alpha' & & \\
 & & B & & & \\
 & \swarrow \text{proj}_1 & \uparrow p & \searrow \text{proj}'_1 & & \\
 & & E & & &
 \end{array} \\
 \\
 & \xrightarrow{p!} & \begin{array}{ccccc}
 & & \exists! \tilde{f} & & \\
 & \text{---} \text{dashed} \text{---} & & \text{---} \text{dashed} \text{---} & \\
 E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2} & E \times_B A' \\
 & \searrow \alpha & \downarrow & \swarrow \alpha' & & \\
 & & B & & & \\
 & \swarrow p(\text{proj}_1) & & \searrow p(\text{proj}'_1) & &
 \end{array}
 \end{array}$$

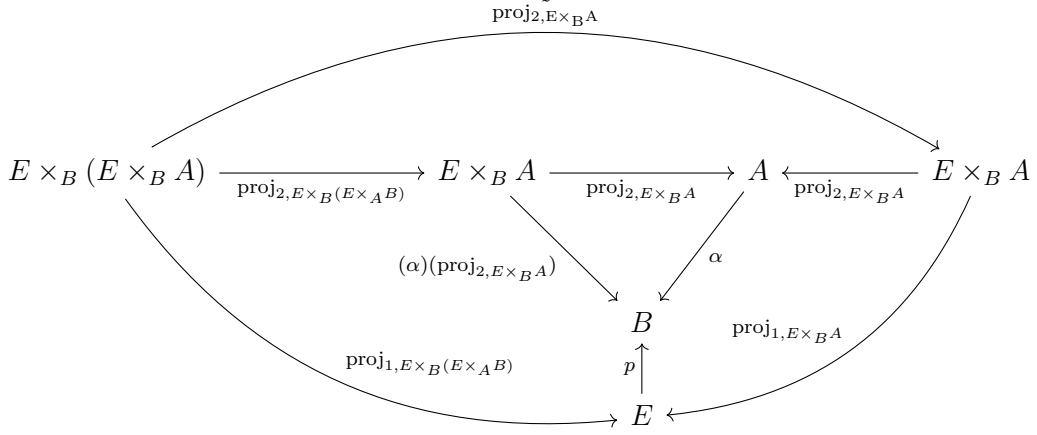
The naturality follow (to complete)

We can check that the unit and counit fulfil the first triangle identities. Let (A, α) be an object in $(\mathcal{A} \downarrow B)$, we have:

$$((p^*\epsilon) \circ (\eta p^*))_{(A,\alpha)} = (p^*\epsilon)_{(A,\alpha)}(\eta p^*)_{(A,\alpha)} = p^*(\text{proj}_{2,E \times_B A})(\eta_{(E \times_B A, \text{proj}_1)}) = \circledast$$

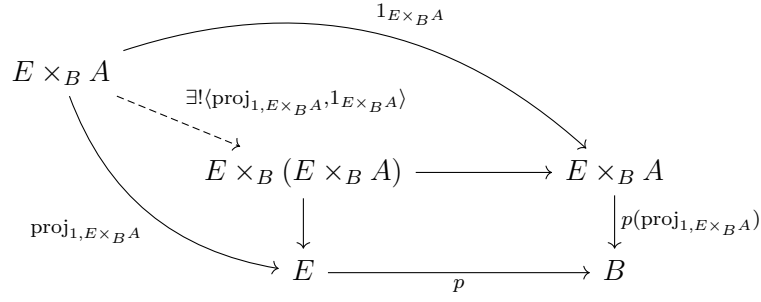
Considering first how p^* acts on the morphism $\text{proj}_{1,E \times_B A}$

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\text{proj}_{2,E \times_B A}} & A \\
 \searrow \alpha(\text{proj}_{2,E \times_B A}) & & \swarrow \alpha \\
 & B &
 \end{array}$$



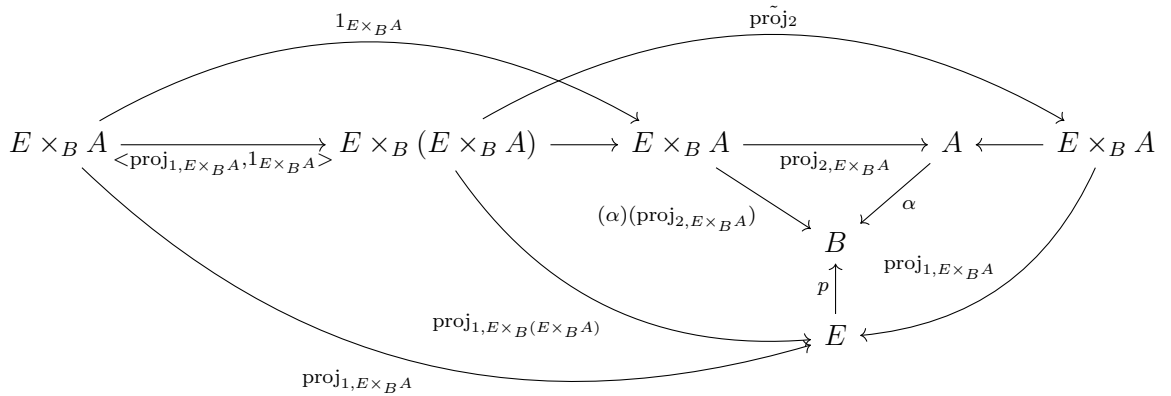
$$\circledast = (\tilde{\text{proj}}_2)(\eta_{(E \times_B A, \text{proj}_1)})$$

and then considering the definition of η , we obtain



$$\circledast = (\tilde{\text{proj}}_2)(\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle)$$

In order to conclude we need to check some compositions so that, by the universal property of the pullback $E \times_B A$ in the right of the following diagram, we obtain the identity morphism.



$$\circledast = (\text{proj}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \text{Id}_{(E \times_B A, \text{proj}_1)}$$

We want to check that composition of the projections of $E \times_B A$ with $\circledast$ is the same of composing those projection with $\text{Id}_{(E \times_B A, \text{proj}_1)}$. Following the first composition

$$\begin{aligned} (\text{proj}_{2, E \times_B A})(\tilde{\text{proj}}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) &= (\text{proj}_{2, E \times_B A})(\text{proj}_{2, E \times_B (E \times_B A)})(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \\ &= (\text{proj}_{2, E \times_B A})(1_{E \times_B A}) = \text{proj}_{2, E \times_B A} = (\text{proj}_{2, E \times_B A})(\text{Id}_{(E \times_B A, \text{proj}_1)}) \end{aligned}$$

and the second composition

$$\begin{aligned} (\text{proj}_{1, E \times_B A})(\tilde{\text{proj}}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) &= (\text{proj}_{1, E \times_B (E \times_B A)})(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \\ \text{proj}_{1, E \times_B A} &= (\text{proj}_{1, E \times_B A})(\text{Id}_{(E \times_B A, \text{proj}_1)}) \end{aligned}$$

Finally we can check the second triangle identities. Let (D, δ) be an object in $(\mathcal{A} \downarrow E)$, we have:

$$\begin{aligned} ((\epsilon_{p!}) \circ (p! \eta))_{(D, \delta)} &= ((\epsilon_{p!})_{(D, \delta)}(p! \eta))_{(D, \delta)} = (\epsilon_{p!(D, \delta)} p!)(\langle \delta, 1_D \rangle) = \\ &= (\epsilon_{(D, p\delta)}) p! (\langle \delta, 1_D \rangle) = \text{proj}_{2, E \times_B D}(\langle \delta, 1_D \rangle) = 1_D = \text{Id}_{(D, \delta)} \end{aligned}$$

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_{2, E \times_B D}} & D \\ \downarrow \text{proj}_{1, E \times_B D} & & \downarrow p\delta \\ E & \xrightarrow{p} & B \end{array}$$

Now we can conclude that

$$\begin{array}{ccc} & \xrightarrow{p^*} & \\ (\mathcal{A} \downarrow B) & \perp & (\mathcal{A} \downarrow E) \\ & \xleftarrow{p!} & \end{array}$$

is an adjunction.

Bibliografia

[1] bla bla bla